CMPT 295

Unit – Microprocessor Design & Instruction Execution

Lecture 29 – Sequential Instruction Execution

Last Lecture

- We have now started to explore how the microprocessor executes machine code instructions (series of 0's and 1's)
- Hardware of a microprocessor (datapath)
  - Made of combinational logic circuits, sequential logic circuits, clock
  - We put combinational logic circuits and sequential logic circuits together
    -> datapath of a microprocessor
- "Constructed" our first microprocessor: executes instructions sequentially

Today's Menu

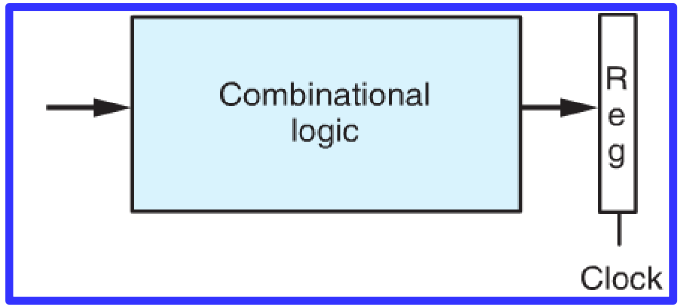
- Instruction Set Architecture (ISA)
  - Definition of ISA
- Instruction Set design
  - Design principles
  - Look at an example of an instruction set: MIPS
  - Create our own
  - ISA evaluation
- Implementation of a microprocessor (CPU) based on an ISA
  - Execution of machine code instructions (datapath)
  - Intro to logic design + Combinational logic + Sequential logic circuit
  - Sequential execution of machine code instructions
  - Staged execution of machine code instructions
  - Pipelined execution of machine code instructions + Hazards

Our first microprocessor: executes instructions sequentially

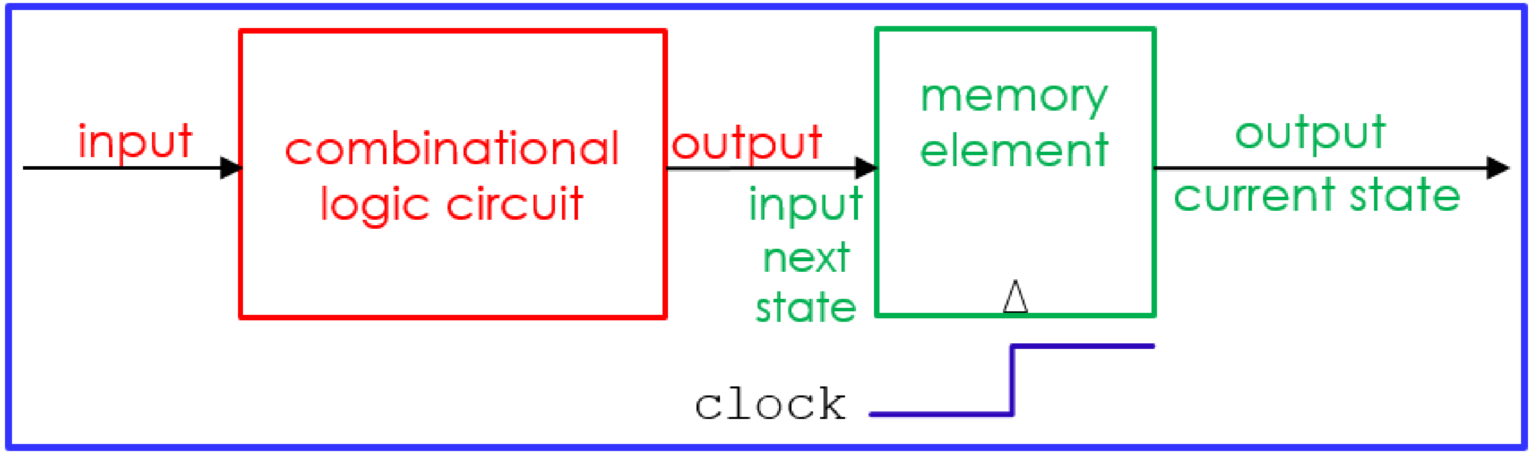
- Now, imagine we have designed and implemented a microprocessor (its datapath):

Called ***sequential instruction execution*** because at every clock cycle, a new machine instruction is executed!

From textbook:



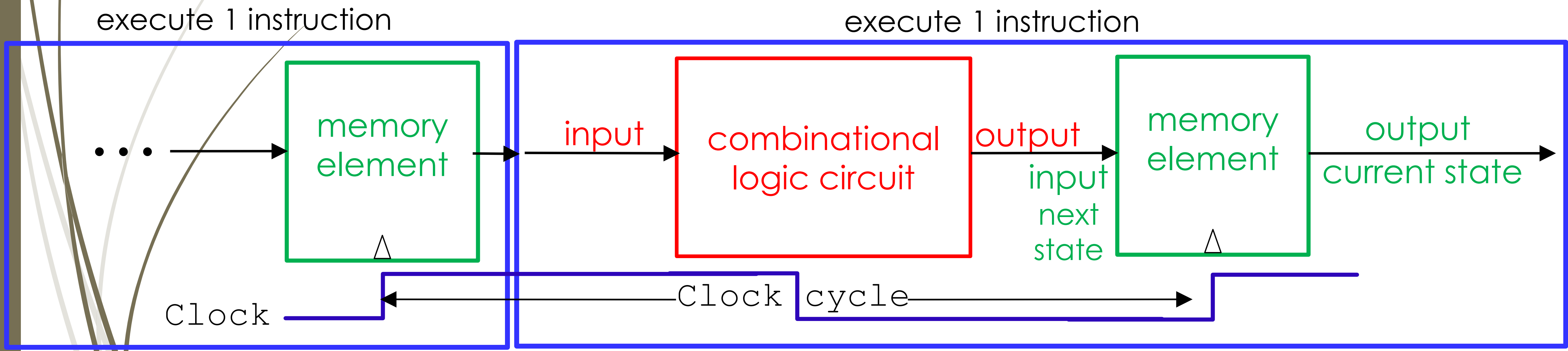
From our lectures:



Analysis of sequential instruction execution

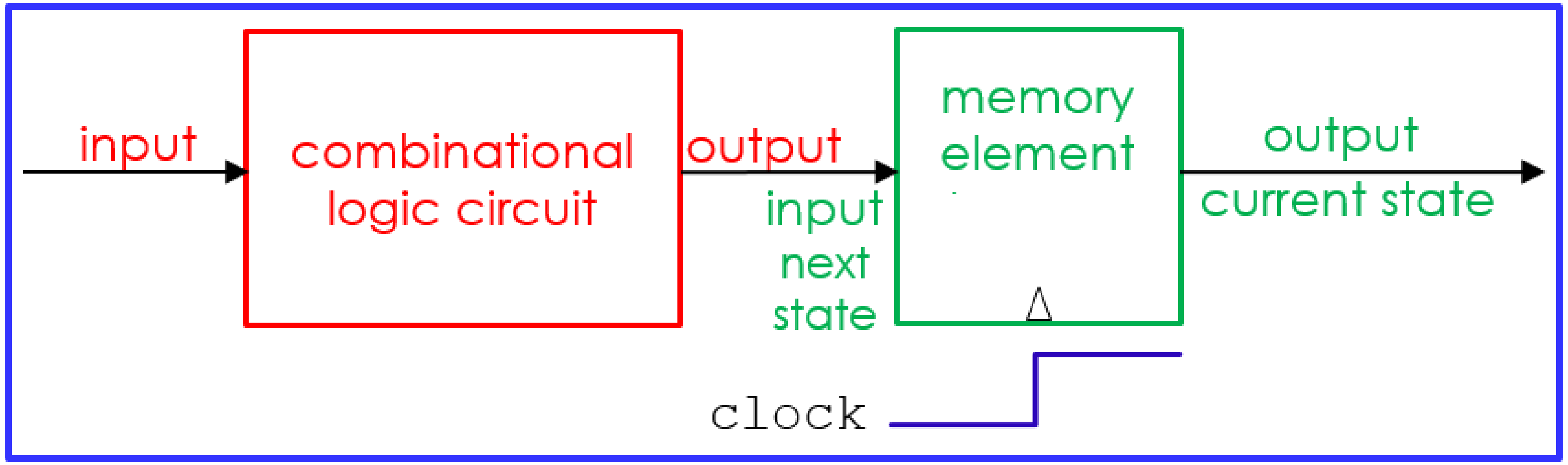
the time between two ***ticks*** of the clock

- Clock cycle required to execute 1 instruction must exceed t_{pd} of combinational logic circuit + t_{pd} of clocked register



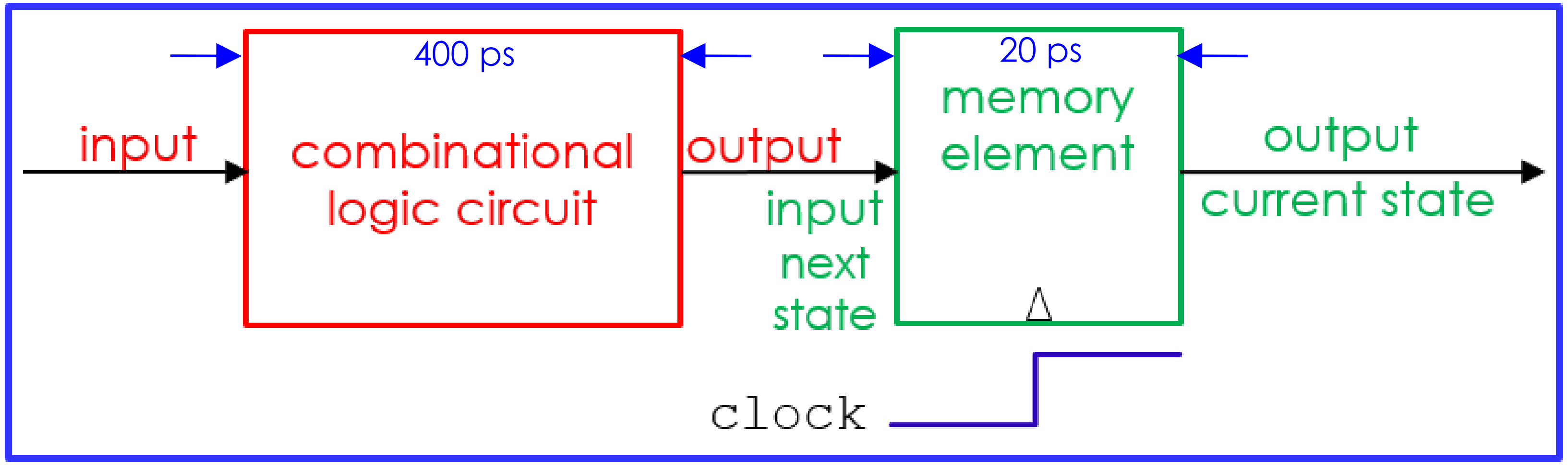
-> t_{pd} + t_{pd} is long clock cycle!

Analysis of sequential instruction execution (cont'd)



- Since the clock cycle is a factor in determining how fast the microprocessor executes machine code instructions (i.e., execution speed), but since our first microprocessor ("sequential instruction execution" microprocessor) needs a **long clock cycle** in order to execute 1 instruction at a time ...
- Then our microprocessor with its long clock cycle is a **slow microprocessor ->** it takes a long time to execute machine code instructions, i.e., to execute a program

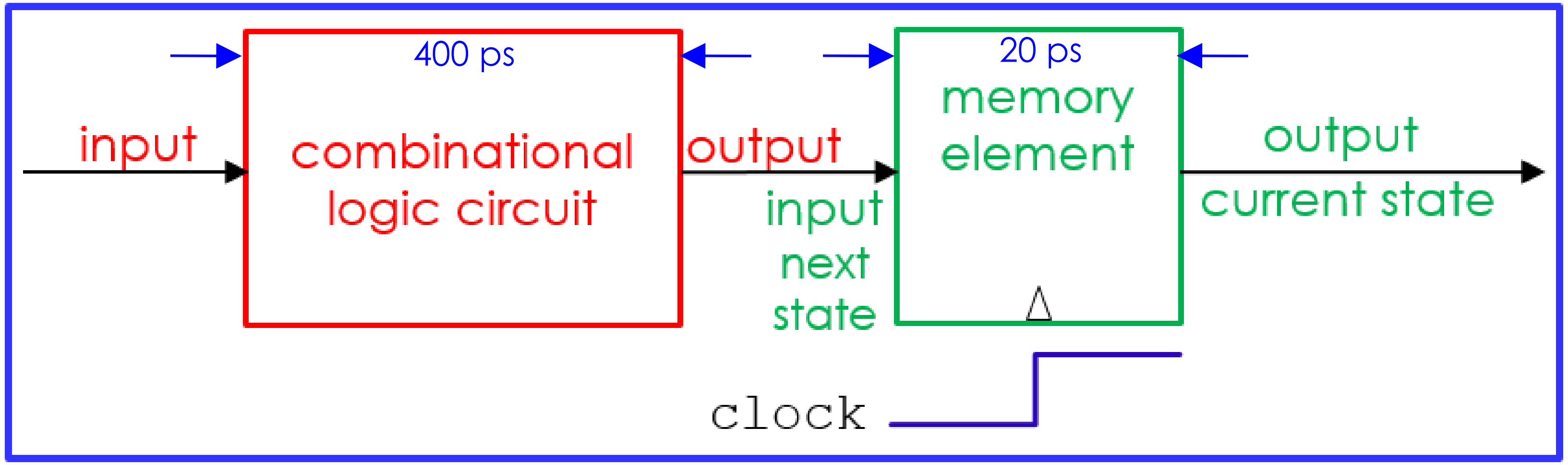
Analysis:



1. **Minimum Clock Cycle**:
   - Def: Amount of time between two “ticks” of the clock

2. **Latency** (a.k.a. propagation delay):
   - Def: Amount of time required to execute a single instruction

Analysis: (cont'd)



3. (instruction) **Throughput**:
   - Def: Number of instructions executed per second - GIPS

4. **CPI**:
   - Def: The number of clock cycles per instruction

Conclusion: sequential instruction execution

- Such microprocessor is called ***single-cycle*** microprocessor
  - Execute each instruction in 1 clock cycle -> CPI = 1
- ***single-cycle*** microprocessor is inefficient:
  - Results in a long clock cycle and a small ***throughput***

Example of a single-cycle microprocessor (datapath)

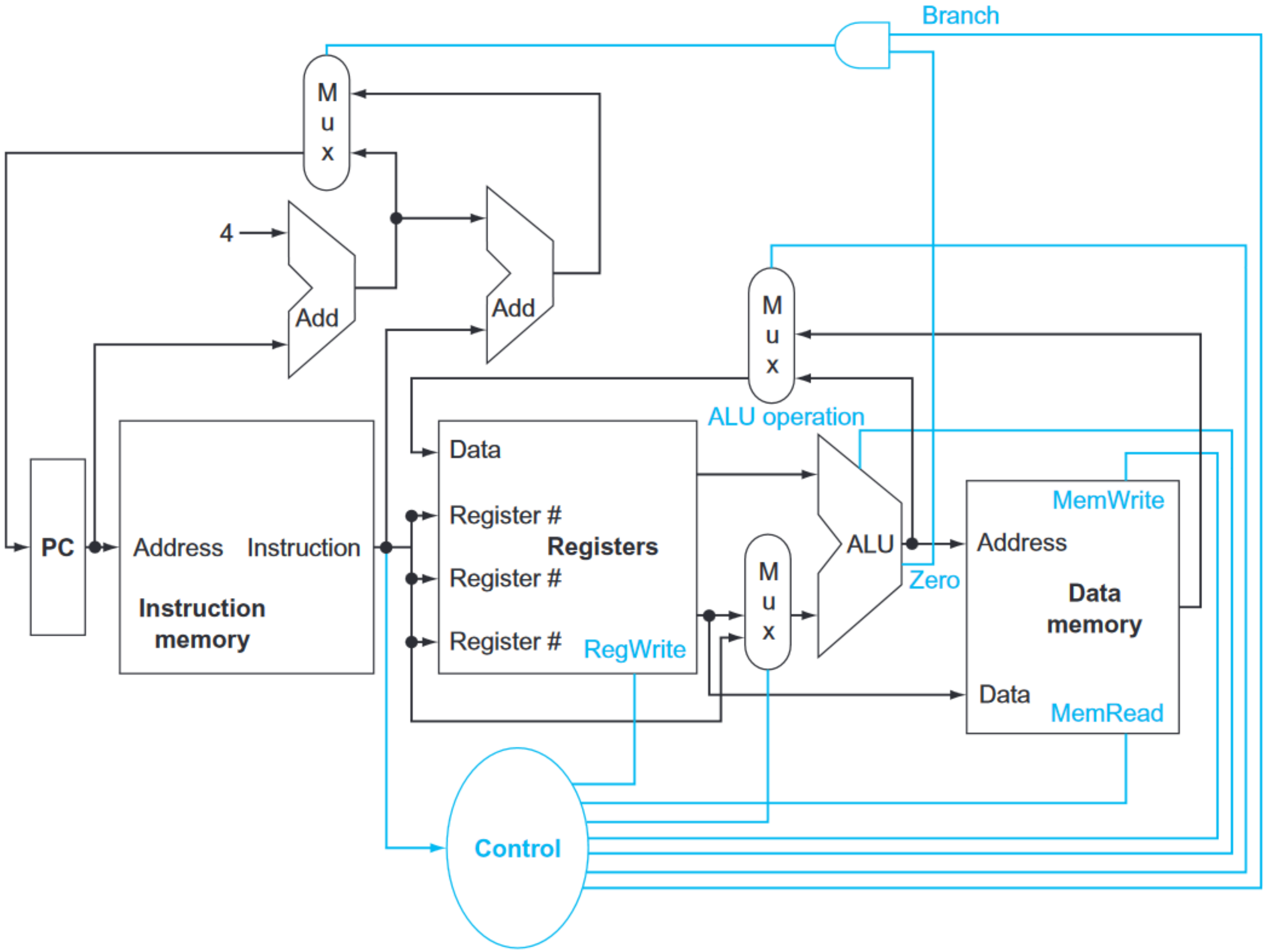


FIGURE 4.2 The basic implementation of the MIPS subset, including the necessary multiplexors and control lines.

https://ict.iitk.ac.in/wp-content/uploads/CS422-Computer-Architecture-ComputerOrganizationAndDesign5thEdition2014.pdf

Summary

- Various models of ***Microprocessor machine instruction execution***:
  - **Model 1: Sequential execution of machine code instructions**
    - The microprocessor we have just *constructed* is a **"sequential execution of machine code instructions"** type of microprocessor since it executes one machine code instruction **at a time** i.e., **per clock cycle**
    - ***Single-cycle*** microprocessor (**CPI = 1**)
- In general, when we analyze various models of ***microprocessor instruction execution***, we look at:
  1. Minimum Clock Cycle: Amount of time between two "ticks" of the clock
  2. Latency: Amount of time required to execute a single instruction
  3. Throughput: Number of instructions executed per second
  4. CPI: The number of clock cycles per instruction

Next Lecture

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